Lecture 02

No bit operations

1 Theorem. The n-th bit can not be extracted in $\mathrm{KC}(2^{\sqrt{n}/a}, \sqrt{n}/a)$, for a large enough constant a.

Proof. Suppose that such a KC-algorithm $\mathcal{A}$ existed for computing the n-th bit in $t(n)$-time using $p(n)$ processors.

Call two integers x and x' t-equivalent if for both x and x' every processor of $\mathcal{A}$ branched the same way up to time t for both inputs x and x'.

How many t-equivalent-classes are there? For given t-equivalence-class C, the content at time t of each memory register r is given by a polynomial in the input $p_r(x)$, of degree at most 2^t. To each branch instruction, e.g., `if` $p_r(x) \leq 0$ `goto` ℓ, corresponds such a register r. Now, p_r has at most 2^t real roots, and if we take all real roots corresponding to all the p_r that appear in some branch instruction at time $t+1$, we have at most $p(n)2^t$ real roots. These roots divide the real line into at most $p(n)2^t + 1$ intervals; if two inputs $x, x' \in C$ belong to the same interval, or if they are equal to the same real root, then every branch instruction branches in the same way for x and x' at time $t+1$. Hence, any t-equivalence-class is divided up into at most $2p(n)2^t + 1$ $(t+1)$-equivalence-classes.

Hence there are at most $p(n)^{O(t(n))}2^{O(t(n)^2)}$ $t(n)$-equivalence classes. Now let us consider all polynomials giving the contents of each register belonging to some branching instruction at some time-step, over all equivalence classes. There are at most $p(n)^{O(t(n))}2^{O(t(n)^2)}$ such polynomials, each having degree at most $2^{t(n}$. To these polynomials correspond at most $p(n)^{O(t(n))}2^{O(t(n)^2)}$ roots, which partition the real line into at most $p(n)^{O(t(n))}2^{O(t(n)^2)}$ intervals. Now, if two integers x, x' belong to the same interval in this partition, the output of $\mathcal{A}$ on x and x' must be the same, hence they must have the same n-th bit. Hence any two n-bit integers x and $x+1$ must belong to different intervals, and so

$$p(n)^{O(t(n))}2^{O(t(n)^2)} \geq 2^n.$$

Now it must follow, for some large enough constant a (depending on the

constants hidden in the O-notation above), that $p(n) \leq 2^{\sqrt{n}/a}$ and $t(n) \leq \sqrt{n}/a$ can not both be true. ■

Parametric complexity

2 *Optimization Problems.* We are looking at numerical problems of the form:

$$\mathcal{A} = \{(\bar{x}, \bar{w}, z) \in \{0,1\}^n \times \mathbb{N}^{n+1} | f(\bar{x}, \bar{w}) \leq z\}$$

for some function $(\bar{x}, \bar{w}) \mapsto f(\bar{x}, \bar{w})$.

2.1. Let us assume that f maps to $\mathbb{N}$ when $\bar{w} \in \mathbb{N}^n$, but is actually defined whenever $\bar{w} \in [0, +\infty)^n$.

2.2. Let us also assume that f is *homogeneous*, meaning that $f(\bar{x}, \alpha\bar{w}) = \alpha f(\bar{x}, \bar{w})$ for every $\alpha \in [0, +\infty)$.

3 *Linear Parametrization.* A *linear parametrization* of cardinality n for $\mathcal{A}$ is a map P_n mapping each real λ in some interval $I_n \subseteq \mathbb{R}$, to an input $P_n(\lambda) \in \{0,1\}^n \times [0, +\infty)^n$ in the domain of f, such that:

3.1. The non-numeric part of $P_n(\lambda)$ is the same for every $\lambda \in I_n$.

3.2. Each numeric parameter in $P_n(\lambda)$ is a linear function of λ with integer coefficients[1].

3.3. The function graph $F(\lambda) = f(P_n(\lambda))$ as λ varies over I_n is piecewise linear. We shall denote the points in this graph by $G(P_n)$.

3.4. The interval I_n is called the *interval of definition.*

3.5. The *bitsize* of P_n, $\beta(P_n)$ is the maximum bitsize among the coefficients of the linear functions of P_n.

3.6. The number of different slopes among the linear segments of $G(P_n)$, denoted $\rho(P_n)$, is called the *complexity* of P_n.

4 *Parametric Complexity.* The parametric complexity of $\mathcal{A}$ for cardinality n is the maximum complexity among all linear-parametrizatiosn of cardinality n for $\mathcal{A}$, i.e.

$$\phi(n) = \max_{P_n} \rho(P_n).$$

4.1. More generally, the parametric complexity of $\mathcal{A}$ for cardinality n and bit-length $\beta(n)$, denoted $\phi(n, \beta(n))$, is the maximum complexity among all P_n with $\beta(P_n) \leq \beta(n)$.

[1] Negative coefficients are OK, but the numeric parameter should remain positive for all $\lambda \in I_n$.

☞ We will show the following theorem in class:

5 Theorem. There exists some constants $a, b \geq 1$ such that no numerical problem with parametric complexity $\phi(n, \beta(n))$ can be solved in $\mathrm{linKC}(2^{\sqrt{\log \phi(n,\beta(n))}/b}, \sqrt{\log \phi(n, \beta(n))}/b)$, even if we only require the algorithm to be correct on inputs whose numerical parameters have bit-length at most $\beta(n)/a$.

☞ Mulmuley's paper shows this without the linear restriction.

Parametric complexity of MaxFlow

6 Theorem. For $f = \mathrm{MaxFlow}$, $\phi(n, n^2) \geq 2^n$.

Proof. Graph was drawn in class.

7 Corollary. MaxFlow cannot be solved in $\mathrm{linKC}(2^{\sqrt{n}/b}, \sqrt{n}/b)$.